Facilitator key and set-up guide

Building the board

1. Print **board-A1.pdf** at 100 % (A1, 841 × 594 mm). Mount it on a steel sheet, or laminate it and fix it to a magnetic whiteboard with magnets. Scale: 14 mm per m³/h, 12.5 mm per m.
2. Print the **3 piece sheets (A1)** (the last one carries the POR/AOR bars, dots and markers) at 100 % on the same printer, laminate, cut along the black outline, and stick magnetic tape along the back of each strip. Alternative: print the pieces on A4/A3 printable magnetic paper, tiled by the print driver at 100 % with cut-and-join marks.
3. Colour code: blue bands are speed curves, sand bands are trimmed impellers, dark blue is two pumps in parallel, brass parabolas are system curves. The thin dark line inside each band is the curve; the band is only there to hold a magnet.
4. Registration: each pump piece carries its Q = 0 end (shut-off head) and white ticks at Q = 10, 20, 30 m³/h. Each system piece is a parabola *relative to the static head*: its bottom-left end sits on the H = H_s grid line at Q = 0. Sliding it up and down is how a team sets the static head — that is the whole point of the piece.
5. POR/AOR bars lie on the Q axis. The green segment is 70–120 % of Q_BEP, amber extends to 50–130 %. There is one bar per pump piece because Q_BEP moves with speed, diameter and the number of pumps.

Running the exercise

1. Teams of 3–4, one board per team (or one board with teams taking turns while the others predict on paper).
2. Card 0 commissions the board. Then hand out A–D one at a time; E and F are for fast teams.
3. Each team registers the pieces, marks BEP, marks its predicted operating point with the red ?, and writes Q, H, % BEP, POR yes/no, η and kW on the card before anyone checks.
4. Verify with the digital board (same pieces, same scenarios, reveals the true intersection) and, for A–D, on the hydraulic loop bench. Score: within 1.5 m³/h and 1.5 m is a hit.
5. Debrief with the answer table below; the discussion points are the learning.

Answer table

Scenario	Pump piece	System	Q m³/h	H m	% Q_BEP	In POR	η %	Shaft kW
0 Commission the board	1480 rpm	H_s 10 m · K 0.020	24.0	21.5	100 %	yes	72	1.95
A Strainer fouls by 40 %	1480 rpm	H_s 10 m · K 0.029	21.4	23.2	89 %	yes	71	1.90
B Supply tank level falls 5 m	1480 rpm	H_s 15 m · K 0.020	20.7	23.6	86 %	yes	71	1.88
C Operator throttles the discharge to 18 m³/h	1480 rpm	H_s 10 m · K 0.047	18.0	25.2	75 %	yes	68	1.83
D VFD reduces speed from 1 480 to 1 200 rpm	1200 rpm	H_s 10 m · K 0.020	16.7	15.6	86 %	yes	71	1.00
E Bonus — second pump started in parallel	2 × parallel ×2	H_s 10 m · K 0.020	29.1 (14.6 each)	27.0	61 %	no — AOR	61	3.52

F Bonus — trim the impeller for 18 m³/h	Ø147	H_s 10 m · K 0.020	18.1	16.5	89 %	yes	71	1.14

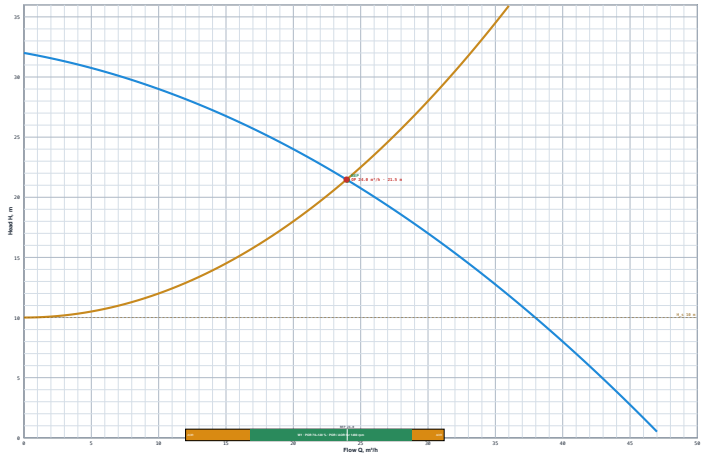
Tolerance for a "hit": ±1.5 m³/h and ±1.5 m. Pieces are 10 mm wide bands; reading the thin centreline gives about ±0.5 m³/h on this scale.

Pieces and their equations

Piece	Description	Equation (Q in m³/h, H in m)	Shut-off m	Q_BEP m³/h
P1	1 480 rpm · Ø174 full impeller	H = (32 − 0.2 q − 0.01 q²), q = Q	32.0	24.0
P2	1 350 rpm	H = 0.832 × (32 − 0.2 q − 0.01 q²), q = Q / 0.912	26.6	21.9
P3	1 200 rpm	H = 0.657 × (32 − 0.2 q − 0.01 q²), q = Q / 0.811	21.0	19.5
P4	1 000 rpm	H = 0.457 × (32 − 0.2 q − 0.01 q²), q = Q / 0.676	14.6	16.2
P5	Ø165 mm trimmed · 1 480 rpm	H = 0.899 × (32 − 0.2 q − 0.01 q²), q = Q / 0.948	28.8	22.8
P6	Ø155 mm trimmed · 1 480 rpm	H = 0.794 × (32 − 0.2 q − 0.01 q²), q = Q / 0.891	25.4	21.4
P7	Ø147 mm trimmed · 1 480 rpm	H = 0.714 × (32 − 0.2 q − 0.01 q²), q = Q / 0.845	22.8	20.3
P8	Two pumps in parallel · 1 480 rpm · Ø174	H = (32 − 0.2 q − 0.01 q²), q = Q/2	32.0	48.0
S1	K = 0.014 (larger bore)	H = H_s + 0.014 Q²	—	—
S2	K = 0.020 baseline · clean strainer · valve open	H = H_s + 0.020 Q²	—	—
S3	K = 0.029 strainer 40 % blocked	H = H_s + 0.029 Q²	—	—
S4	K = 0.047 discharge throttled	H = H_s + 0.047 Q²	—	—
S5	K = 0.070 heavily throttled	H = H_s + 0.070 Q²	—	—

Strainer fouled 40 %: loss through a screen scales with 1/(open area)², so the strainer's 0.005 becomes 0.005/0.6² = 0.0139 and K = 0.015 + 0.0139 = 0.029. Throttling to 18 m³/h: K = (H_pump(18) − H_s)/18² = 0.0468. Trim for 18 m³/h: diameter ratio 0.843 → Ø147 mm.

0 · Commission the board



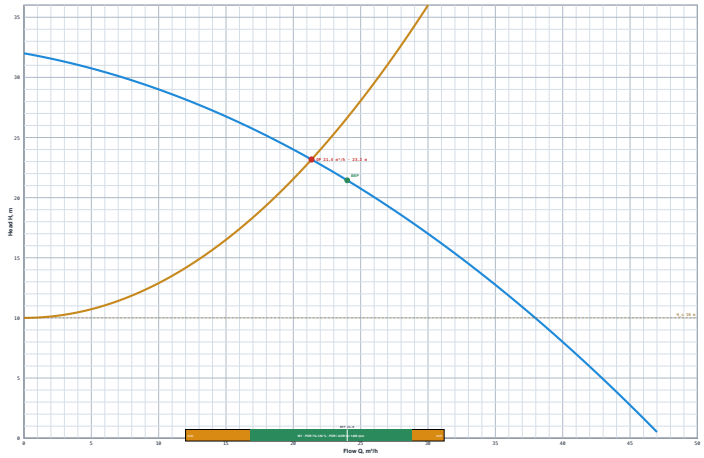
Set-up: Pump at 1 480 rpm with the full Ø174 impeller. Supply tank level gives a static head of 10 m; clean strainer, discharge valve fully open (K = 0.020). Place the pump curve, the system curve and the POR/AOR bar, mark the BEP, and mark the operating point.

Answer: The curves cross at 24.0 m³/h and 21.5 m — right on the BEP (100 % of Q_BEP), efficiency 72 %, shaft power 1.95 kW. The board is now the reference for every scenario.

Q m³/h	H m	% Q_BEP	η %	kW
24.0	21.5	100	72	1.95

Watch for: Teams that place the system piece with its end at H = 0 instead of the static head. Teams that mark BEP at the top of the curve.

A · Strainer fouls by 40 %



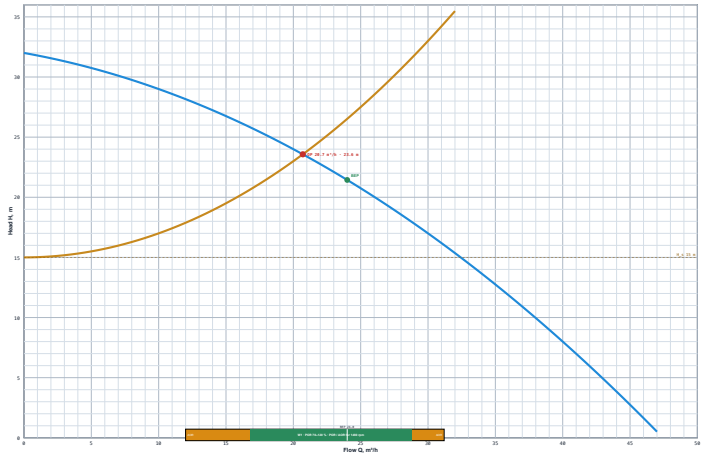
Set-up: Debris blocks 40 % of the suction strainer area. Nothing else changes. Loss through a screen goes with 1/(open area)², so the strainer coefficient rises by (1/0.6)² = 2.8×: K goes from 0.020 to 0.029.

Answer: Same static head, steeper parabola: the operating point slides up and left along the pump curve. Flow falls, head rises, the pump moves left of BEP but stays inside the POR. The loss is on the suction side, so NPSH margin falls — the cavitation trainer shows what comes next.

Q m³/h	H m	% Q_BEP	η %	kW
21.4	23.2	89	71	1.90

Watch for: Teams that move the static head instead of changing K. Ask where the loss is: suction side, so NPSH margin falls.

B · Supply tank level falls 5 m



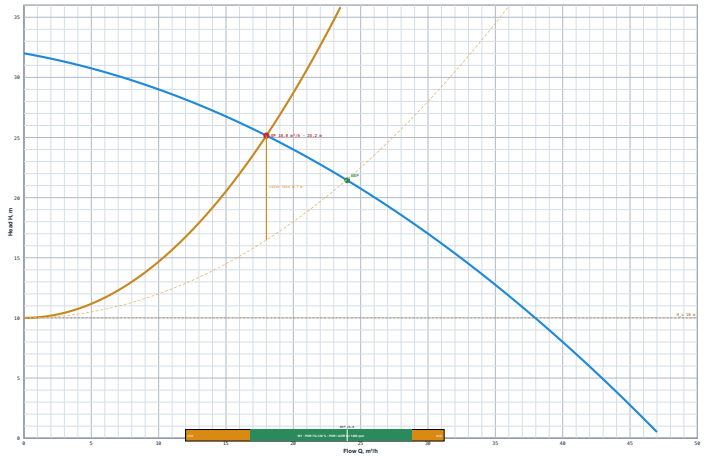
Set-up: The supply tank is being drawn down and its level drops 5 m. The pump lifts from the tank to the same delivery point, so the static head rises from 10 m to 15 m. Friction unchanged (K = 0.020).

Answer: The parabola keeps its shape and moves up 5 m. The operating point moves up and left along the pump curve; the whole curve shift costs more flow than the strainer did. Static head is the one thing a bigger valve opening cannot buy back.

Q m³/h	H m	% Q_BEP	η %	kW
20.7	23.6	86	71	1.88

Watch for: Teams that slide the piece down (level falls, head rises — the pump lifts further). Teams that change the shape of the parabola.

C · Operator throttles the discharge to 18 m³/h



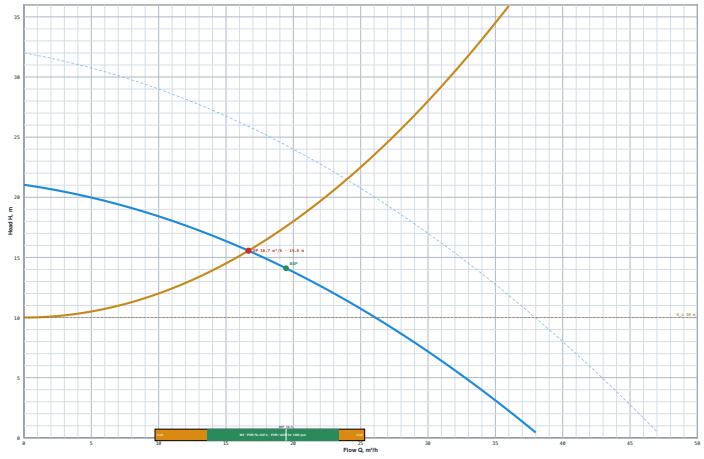
Set-up: Production asks for 18 m³/h instead of 24. The operator closes the discharge valve until the flowmeter reads 18 m³/h. Static head 10 m; the extra loss is all in the valve.

Answer: The pump curve at 18 m³/h reads 25.2 m, so the system parabola must pass through that point: K ≈ 0.047 (piece S4). The valve is destroying 8.7 m of head — 34 % of what the pump produces. Shaft power 1.83 kW.

Q m³/h	H m	% Q_BEP	η %	kW
18.0	25.2	75	68	1.83

Watch for: Teams that pick S3 because "it is steeper": ask them to check where it crosses. Make them draw the valve loss as a vertical distance at 18 m³/h.

D · VFD reduces speed from 1 480 to 1 200 rpm



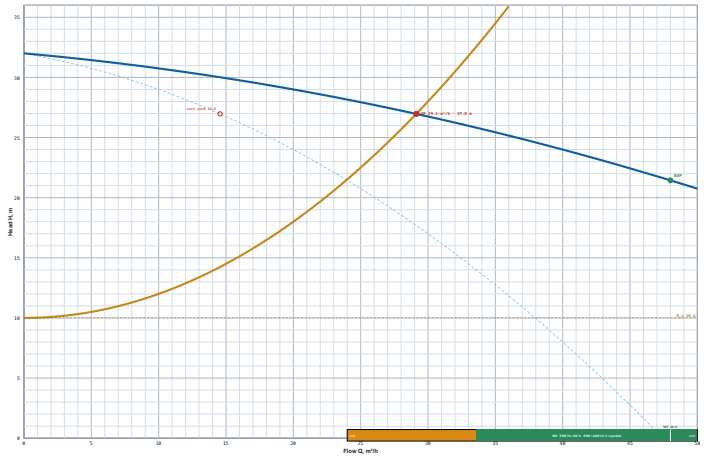
Set-up: Instead of throttling, the same flow reduction is attempted with the VFD. Speed goes to 1 200 rpm; the valve stays fully open (K = 0.020, static head 10 m).

Answer: Flow scales with speed but head with speed², so the curve shrinks toward the origin and the operating point lands at about 16.7 m³/h and 15.6 m. The BEP moves to 19.5 m³/h, so the pump still runs at 86 % of BEP flow — inside the POR. Shaft power 1.00 kW against 1.80 kW if the same flow were throttled: the static head keeps the saving below the cube law.

Q m³/h	H m	% Q_BEP	η %	kW
16.7	15.6	86	71	1.00

Watch for: Teams that leave the POR bar and BEP where they were. Teams that expect the cube law ($P \propto N^3$) to hold: the static head breaks it.

E · Bonus — second pump started in parallel



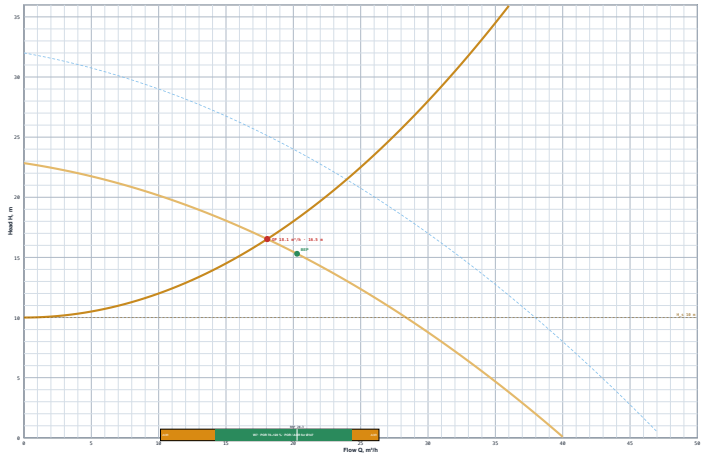
Set-up: A second identical pump is started in parallel to "double the flow". Baseline system (static head 10 m, K = 0.020), both at 1 480 rpm.

Answer: Total flow rises only to 29.1 m³/h, 22 % more, not double, because the system curve is steep. Each pump delivers 14.6 m³/h — 61 % of BEP, left of the POR. Two pumps in parallel only pay off on flat (friction-light) systems.

Q m³/h	H m	% Q_BEP	η %	kW
29.1 (14.6 per pump)	27.0	61	61	3.52

Watch for: Teams that draw the parallel curve by adding heads. Teams that expect 48 m³/h. Ask what each pump sees: 61 % of BEP, in the low-flow recirculation zone.

F · Bonus — trim the impeller for 18 m³/h



Set-up: The 18 m³/h duty is permanent. Instead of throttling (C) or a VFD (D), the impeller is trimmed. Baseline system, 1 480 rpm.

Answer: A diameter ratio of 0.843 (Ø147 mm) gives 18 m³/h; the Ø147 piece is the nearest. Operating point about 18.1 m³/h and 16.5 m at 1.14 kW. A 16 % trim is beyond the usual 10–15 % limit where efficiency starts to suffer; the affinity laws for diameter are only reliable for small trims.

Q m³/h	H m	% Q_BEP	η %	kW
18.1	16.5	89	71	1.14

Watch for: Teams that scale only head or only flow. Point out the trim limit and that the trimmed pump cannot be "un-trimmed" when the duty comes back.

Verify on the hydraulic loop bench

The loop bench simulator uses the same pump. Card A ≈ strainer fouling slider; card B ≈ reservoir level; card C ≈ discharge valve; card D ≈ speed 81 %. Its pipe friction differs from the board's K, so compare the *direction and size* of each move, not the exact numbers.

[Illustrative pump and system data, for training only.] Generated from model.js — regenerate with node make-print.js after any change.